

United International University
School of Science and Engineering

Trimester: Summer 2025

Course Title: Calculus and Linear Algebra

Course Code: Math 183/ Math 2183

Assignment-1

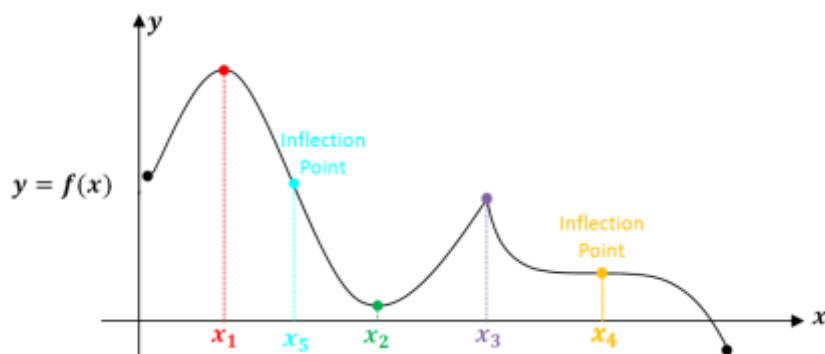
Topic : Analysis of function, Partial derivative and Chain rule

1. Consider the function $f(x) = x^5 - x^3$
 - (a) Find the intercepts of the function.
 - (b) Find all the critical and stationary points of the function.
 - (c) Find the intervals on which $f(x)$ is increasing or decreasing.
 - (d) Find the interval on which $f(x)$ is concave up and down and hence find the inflection points.
 - (e) Find the relative extrema of $f(x)$ using both first and second derivative test.
 - (f) Make a conjecture about the behavior of the graph of $f(x)$ in the vicinity of its x – intercepts. Also discuss the end behaviour of $f(x)$.
 - (g) Use the above results to draw the graph of the given function.

2. A sign chart is presented for the first and second derivatives of a function f . Assuming that f is continuous everywhere, find: (a) the intervals on which f is increasing, (b) the intervals on which f is decreasing, (c) the open intervals on which f is concave up, (d) the open intervals on which f is concave down, and (e) the x -coordinates of all inflection points.

INTERVAL	SIGN OF $f'(x)$	SIGN OF $f''(x)$
$x < 1$	–	+
$1 < x < 2$	+	+
$2 < x < 3$	+	–
$3 < x < 4$	–	–
$4 < x$	–	+

3.



- i. Find the sign of $f'(x)$ at the point x_1, x_2, x_3, x_4 and x_5
 - ii. Find the sign of $f''(x)$ at the point x_1, x_2, x_3, x_4 and x_5
 - iii. Which points are considered as stationary point?
4. Find all the critical points and stationary points for each of the following function
- (a) $f(x) = x^{2/3}(6 - x)^{1/3}$
 - (b) $f(x) = x^{3/5}(4 - x)$
5. A rectangular field is to be bounded by a fence on three sides and by a straight stream on the fourth side. Find the dimensions of the field with maximum area that can be enclosed using 1000 ft of fence.
6. A garden is to be laid out in a rectangular area and protected by a chicken wire fence. What is the largest possible area of the garden if only 200 running feet of chicken wire is available for the fence?
7. Find the slope of the surface $z = f(x, y) = \ln(xy^2 + 2xy) + xy$ in the x direction at the point $(2, -1)$.
8. Let $z = \cos(x - 2y)$, where $x = uv$, $y = u - v$. Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.
9. Use implicit differentiation to find $\frac{dy}{dx}$, where $5x^4 + y^3 + e^{xy} = 0$
10. The population of a community is known to increase at a rate proportional to the number of people present at time t . If an initial population P_0 has doubled in 5 years, how long will it take to triple? To quadruple?
11. The radioactive isotope of lead, Pb-209, decays at a rate proportional to the amount present at time t and has a half-life of 3.3 hours. If 1 gram of this isotope is present initially, how long will it take for 90% of the lead to decay?

Date of submission:

